



Buying Tips

✓ BACKUP OR RESTORATION

UTILITIES: If you don't mind re-installing your OS in the event of a crash and then copying all the backup files to the PC, opt for backup software. On the other hand, restoration software do their job without any manual intervention and OS re-installation.

✓ **TYPE OF BACKUP:** Choose software that let you perform an incremental backup, since this feature saves a lot of space and time.

✓ **SIZE OF BACKUP DATA:** Backup software that let you compress data save you a lot of hard disk space in the long run.

✓ **NETWORK BACKUP:** This feature lets you maintain a backup over the network. Data is stored at another location and can be retrieved with ease.

✓ **EASE OF USE:** Opt for a program that has a user-friendly interface, especially if it's to be used with the family PC.

✓ **SCHEDULED BACKUP:** Manual backups can get tiresome and lead to disastrous consequences if you forget to take them on time. Hence, opt for software that lets you schedule backups at fixed intervals.

✓ **MANUAL OR DOCUMENTATION:** See to it that the backup or restoration software comes with adequate documentation. This come in handy when, for example, you need to perform a restore and don't know what exactly to do.

Quick Find

YOU NEED A way to bring your system back to its original state with minimal trouble

Basic Use

LOOK FOR A system-restoration software that has a user-friendly interface, and a decent number of features.

YOU NEED To secure your data by backing it up and copying it to another location.

Inter Use

LOOK FOR A feature-rich backup solution that supports complete as well as incremental backups, locally and across networks.

Data Destroyers

Data loss can occur due to a number of reasons. Some of the common ones are listed below:

Virus attack: Many viruses are notorious for deleting files. Some try to delete or overwrite the partition table, which stores the start and end locations of the data partitions on the disk.

Accidental deletion of files: This is a very simple reason for data loss—the user deletes some important files without moving them to the recycle bin first. The user may also delete files from the command prompt, or through applications that do not utilise the Recycle Bin, for example, WinRar.

Physical problems with the hard disk: This is by far the most common reason for data loss, and also the most difficult to remedy. Hard disks can develop a multitude of problems, starting from faulty read and write head alignment, to a badly damaged head actually glued to the hard disk platters due to stiction (static friction).

Precautions to be taken in the event of data loss:

Data loss always seems to happen at the wrong times, but a few simple precautions can lead to a much better chance of recovering data.

First of all, make sure you don't shut down or restart the machine. Don't install anything on the damaged drive—this also includes data recovery software. If the recovery software isn't already installed, run it from a CD or floppy. Remember, the lesser the disk activity there is after the crash, the more data you're likely to recover.

Backup strategies

Backup strategies change from user to user. One of the primary decisions is the frequency of backup. For home users, a manual backup as and when required ought to be enough, but for small firms with about 10 to 20 employees, this won't work. Here, a systematic schedule must be set up to make sure that data is backed-up regularly.

Do remember

- Remember to use anti-virus software, especially if you're used to downloading from the Internet. To prevent your data from being corrupted, update your anti-virus program as often as possible.
- Prevent your storage devices from impacts and shocks, especially when they are running. Don't move your cabinet about when the power is on.
- Avoid electro-static discharges when handling and installing your drives. Do not touch electric contacts on the controller board with bare hands. When handling devices, make sure your other hand is touching something metallic such as the cabinet—this will earth you and prevent electrostatic discharges.
- Hard drives should be regularly defragmented, and file systems checked. Any software or data that are no longer in use should be uninstalled or removed.

Facts

- Backup software just pick up data from a source and dump it into another location. Restoration software take periodic snapshots of the entire hard disk—including the OS files—that can be used at any later date to bring the computer to the same state as before.
- As far as possible, schedule backups to happen after office hours.

- All restoration utilities use their own environments—generally customised operating systems—during the restore operations. Thus, they can restore the PC to its former state even if the OS is corrupted.

Crash

- Backups can be complete or incremental; a complete backup will copy all the files in the folder irrespective of whether the files are modified or not, whereas incremental backups will only copy those files that are modified.

The Future

It is now possible to recover almost all data from a hard disk without resorting to special tools that scavenge your hard disk for magnetic patterns. All you need to do is take periodic backups using automated tools. Alternatively, you may use rollback software that are capable of restoring a corrupted OS to its former state. With the need to salvage and backup data becoming increasingly apparent even amongst home users, this particular segment can expect to see some interesting developments in the next few years.